WHMIS 2015

The Basics



WorkSafeBC

About WorkSafeBC

At WorkSafeBC, we're dedicated to promoting safe and healthy workplaces across B.C. We partner with workers and employers to save lives and prevent injury, disease, and disability. When work-related injuries or diseases occur, we provide compensation and support injured workers in their recovery, rehabilitation, and safe return to work. We also provide no-fault insurance and work diligently to sustain our workers' compensation system for today and future generations. We're honoured to serve the workers and employers in our province.

Prevention Information Line

We provide information and assistance with health and safety issues in the workplace.

Call the information line 24 hours a day, 7 days a week to report unsafe working conditions, a serious incident, or a major chemical release. Your call can be made anonymously. We can provide assistance in almost any language.

If you have questions about workplace health and safety or the Occupational Health and Safety Regulation, call during our office hours (8:05 a.m. to 4:30 p.m.) to speak to a WorkSafeBC officer.

If you're in the Lower Mainland, call 604.276.3100. Elsewhere in Canada, call toll-free at 1.888.621.7233 (621.SAFE).

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WorkSafeB

Health and safety resources

You can find our health and safety resources on worksafebc.com, and many of them can be ordered from the WorkSafeBC Store at worksafebcstore.com.

In addition to books, you'll find other types of resources at the WorkSafeBC Store, including DVDs, posters, and brochures. If you have any questions about placing an order online, please contact a customer service representative at 604.232.9704 or toll-free at 1.866.319.9704.

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WorkSafeBC wishes to thank the Canadian Centre for Occupational Health and Safety (CCOHS) for permission to use its WHMIS educational materials in this publication.

Introduction

This book explains the basics of the Workplace Hazardous Materials Information System (WHMIS), including the following:

- What is WHMIS? — background to WHMIS; key changes from WHMIS 1988 to WHMIS 2015; key elements and participants
- Classification — hazard groups, classes, and categories; products not covered by WHMIS
- Pictograms — introduction; pictograms vs. WHMIS 1988 hazard symbols; how pictograms are used and where to find them
- Labels — supplier labels, workplace labels, and other identifiers
- Safety data sheets (SDSs) — use, content, format, and trade secrets
- Putting WHMIS into action — WHMIS program; education and training

The overall purpose of WHMIS is to help ensure a safer, healthier workplace. WHMIS education helps you to understand how WHMIS works. WHMIS training gives you hands-on knowledge of how to work safely with specific products at your workplace.

When you have been successfully educated and trained in WHMIS, you should be able to answer these four questions:



- What are the hazards associated with hazardous products? (For example, "How can this product hurt me?")
- How do I protect myself? (For example, "What personal protective equipment should I wear?")
- What should I do in an emergency? (For example, "What do I do if I spill it?")
- Where do I get more information?

By understanding the information in this book, you will learn where to look on the label of a hazardous product, where to look on a safety data sheet, and what to discuss with your supervisor.

To help you find out how much you've learned, several exercises are included near the end of this book.

1 What is WHMIS?

Overview of WHMIS

WHMIS provides you with health and safety information about hazardous products in your workplace.

Exposure to hazardous products can result in health problems such as irritation of the eyes, sensitization of the skin or lungs, heart ailments, kidney and lung damage, or cancer. Hazardous products can cause fires, explosions, or other accidents when improperly stored or handled.

Under WHMIS, you have the right to receive information about each hazardous product you use, handle, or store (for example, its identity, its hazards, and the safety precautions you need to take). You can use this information to help make sure you go home alive and well at the end of every workday.

What does WHMIS stand for?

- **Workplace**
  - Deals only with products used in the workplace
- **Hazardous Materials**
  - Dangerous products that may cause fires, explosions, or health problems
- **Information System**
  - Provides information about hazardous products



WHMIS gets an update

WHMIS first came into effect in 1988 across Canada. It was updated in early 2015 to reflect a new set of rules called the Globally Harmonized System of Classification and Labelling of Chemicals (GHS).

Developed by the United Nations, GHS does the following:

- Defines and classifies the hazards of chemical products
- Provides health and safety information on labels and safety data sheets, or SDSs (previously called material safety data sheets or MSDSs in the original WHMIS)

GHS is a worldwide system. Its goals are for the whole world to adopt and use:

- The same set of rules for classifying hazardous products
- The same format and content for labels and SDSs

Currently, many countries have different systems for classifying and labelling hazardous products. In fact, several different systems can exist even within the same country. This situation has been confusing for everyone.

GHS has not replaced WHMIS. Instead, GHS introduces some important changes to WHMIS. This will result in many benefits, such as the following:

- Providing improved, consistent hazard information
- Encouraging the safe handling and use of hazardous products
- Promoting better emergency response
- Making it easier and less expensive for companies to follow the rules
- Making trade easier
- Reducing the costs of regulation and enforcement

For the sake of clarity, the original WHMIS is now called WHMIS 1988. The updated version is called WHMIS 2015.

Key changes from WHMIS 1988 to WHMIS 2015 include the following:

- The term “hazardous product” replaces “controlled product.”
- Hazard classification criteria are more complete. This improves the ability to show the severity of hazards.
- New hazard classes are included (for example, “Aspiration hazard”).
- The language has been standardized (made more consistent).
- Supplier labels have a few new requirements (for example, the use of specific signal words, hazard statements, pictograms, and precautionary statements).
- SDSs follow a 16-section format, and the information in those sections has been standardized.

By late 2018, all suppliers and employers were required to comply with WHMIS 2015. WHMIS 1988 no longer applies.

WHMIS legislation

WHMIS is based on a series of laws and regulations passed by the federal, provincial, and territorial governments.

Federal legislation (the *Hazardous Products Act* and Hazardous Products Regulations) deals with the importation and sale of hazardous products. Health Canada reviews claims for confidential business information ("trade secrets").

Provincial legislation (for example, B.C.'s *Workers Compensation Act* and Occupational Health and Safety Regulation) covers the use of hazardous products in the workplace.

WHMIS in brief

- A nationwide system put in place at the federal, provincial, and territorial levels in 1988; updated in 2015
- Recognizes the interests of all concerned
  - Labour (workers)
  - Owners (employers)
  - Industry (suppliers)
  - Government (regulators)
- Four key elements of WHMIS 2015
  - Classification
  - Labels
  - Safety data sheets (SDSs)
  - Worker education and training

Key elements of WHMIS 2015

- **Classification**
  Hazardous products are classified by the types of hazards they present. WHMIS 2015 divides hazardous products into two **hazard groups**: physical hazards and health hazards. The two hazard groups are further divided into **hazard classes**. For more information, see page 12.

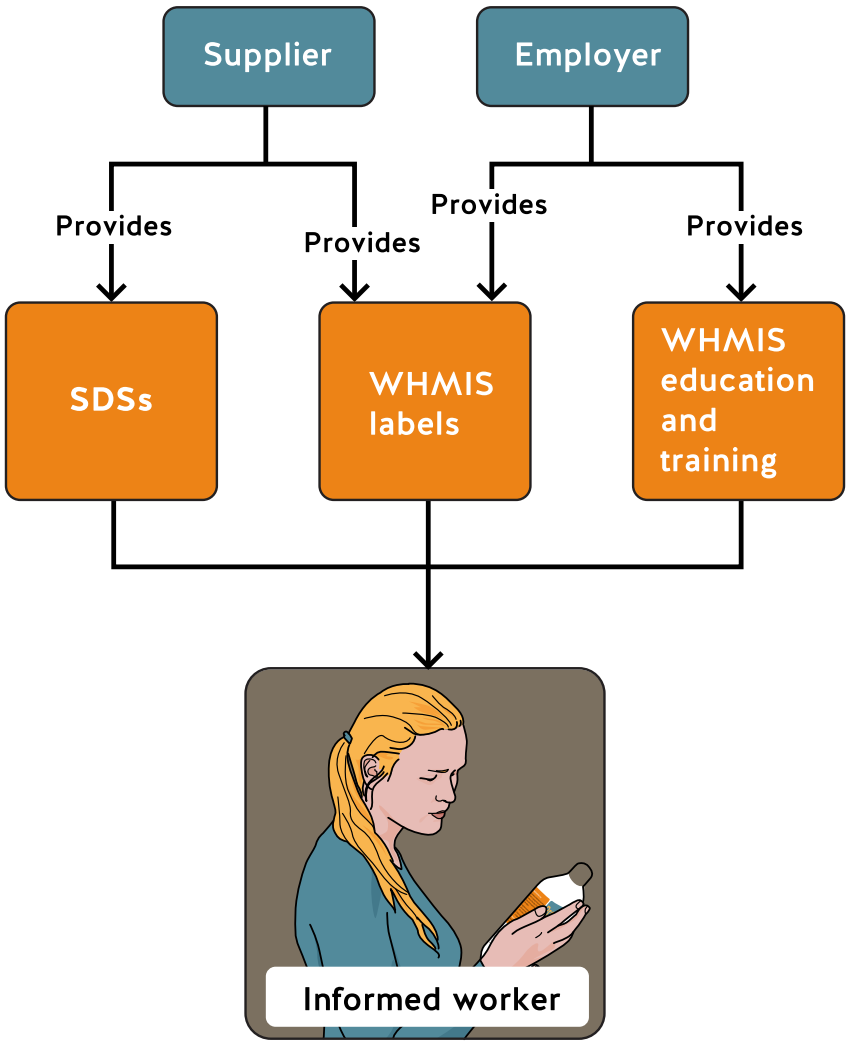
  After a hazardous product has been classified, the following three WHMIS elements are used to communicate health and safety information.

- **WHMIS labels**
  Labels on hazardous products alert you to the identities of products, their hazards, and the precautions you'll need to take. The information on hazards and precautions has been standardized.

- **Safety data sheets (SDSs)**
  These documents provide detailed hazard and precautionary information. Under WHMIS 2015, SDSs use a 16-section format. The information required in each section has been standardized.

- **WHMIS education and training programs**
  Your employer provides education and training for you so you can work safely with or near hazardous products. As a worker, you need to know:
  - How WHMIS works
  - The hazards of hazardous products in your workplace
  - The safe work procedures you must follow

How does WHMIS 2015 work?



Key WHMIS participants and their responsibilities

As a **worker**, you are one of the key participants in WHMIS, along with **employers** and **suppliers**. (Suppliers are organizations or individuals who make, import, sell, or distribute hazardous products in Canada.)

All three groups — workers, employers, and suppliers — have specific responsibilities, as shown in the following table. The roles and responsibilities of each group remain unchanged in WHMIS 2015.

Table 1. Group responsibilities

Workers
• Take part in WHMIS training programs. • Take necessary steps to protect yourself and your co-workers. • Take part in identifying and controlling hazards.
Employers
• Educate and train workers on the hazards and safe use of hazardous products. • Ensure that hazardous products are properly labelled. • Prepare workplace labels and SDSs (as needed). • Ensure that up-to-date SDSs are easily available to workers. • Ensure effective control measures are in place to protect workers.
Suppliers
• Properly classify all hazardous products. • Obtain or prepare up-to-date labels and SDSs. • Provide these labels and SDSs to purchasers of hazardous products.

2 Classification of hazardous products

How hazardous products are classified

What's new

- New classification criteria
- Two hazard groups (physical hazards, health hazards)
- 19 physical hazard classes
- 12 health hazard classes
- Hazard classes containing "categories" or "types" that reflect varying degrees of hazard

The way that hazardous products are classified has changed in WHMIS 2015. Hazardous products are now divided into two **hazard groups**:

- **Physical hazards**, based on the physical or chemical properties of the product (for example, products that are flammable, reactive, or corrosive to metals)
- **Health hazards**, based on the ability of the product to cause a health effect, such as:
  - Eye irritation
  - Respiratory sensitization (may cause allergy or asthma symptoms, or breathing difficulties)
  - Carcinogenicity (may cause cancer)

The two hazard groups are further divided into **hazard classes**. A brief listing of the hazard classes in each hazard group is shown below, followed by an overview of the classes. Hazard classes are a way of grouping together products that have similar properties.

Physical hazards

The physical hazards group includes the following hazard classes:

- Combustible dusts
- Corrosive to metals
- Flammable aerosols
- Flammable gases
- Flammable liquids
- Flammable solids
- Gases under pressure
- Organic peroxides
- Oxidizing gases
- Oxidizing liquids
- Oxidizing solids
- Pyrophoric gases
- Pyrophoric liquids
- Pyrophoric solids
- Self-heating substances and mixtures
- Self-reactive substances and mixtures
- Simple asphyxiants
- Substances and mixtures which, in contact with water, emit flammable gases
- Physical hazards not otherwise classified

Note

Explosives are classified as physical hazards under GHS. And the "exploding bomb" pictogram appears in WHMIS 2015 because some hazardous products can explode. But explosives are not included in WHMIS 2015 because other laws cover them.

Health hazards

The health hazards group includes the following hazard classes:

- Acute toxicity
- Aspiration hazard
- Biohazardous infectious materials
- Carcinogenicity
- Germ cell mutagenicity
- Reproductive toxicity
- Respiratory or skin sensitization
- Serious eye damage/eye irritation
- Skin corrosion/irritation
- Specific target organ toxicity — single exposure
- Specific target organ toxicity — repeated exposure
- Health hazards not otherwise classified

Note

GHS includes an environmental hazards group. This group was not adopted in WHMIS 2015 because it is beyond the scope of WHMIS (i.e., workplaces). However, you may see the environmental classes listed on labels and SDSs, and this is allowed under WHMIS 2015.

Hazard classes

A hazardous product is a product that falls into one or more of the hazard classes described below. Suppliers classify these products and assign one or more pictograms. (Pictograms are symbols surrounded by borders; see page 20.)

Brief descriptions of each of the hazard classes are listed on the following pages.

Table 2. Overview of physical hazard classes

Physical hazard class	General description
Flammable gases **Flammable aerosols** **Flammable liquids** **Flammable solids**	These four classes cover products that have the ability to ignite (catch fire) easily. The main hazards are fire or explosion.
Oxidizing gases **Oxidizing liquids** **Oxidizing solids**	These three classes cover oxidizers, which may cause or intensify a fire or cause a fire or explosion.
Gases under pressure	This class includes compressed gases, liquefied gases, dissolved gases, and refrigerated liquefied gases. Compressed gases, liquefied gases, and dissolved gases are hazardous because of the high pressure inside the cylinder or container. The cylinder or container may explode if heated. Refrigerated liquefied gases are very cold. They can cause severe cold (cryogenic) burns or injury.
Self-reactive substances and mixtures	These products may react on their own to cause a fire or explosion, or may cause a fire or explosion if heated.
Pyrophoric liquids **Pyrophoric solids** **Pyrophoric gases**	These products can catch fire very quickly (spontaneously) if exposed to air.
Self-heating substances and mixtures	These products may catch fire if exposed to air. These products differ from pyrophoric liquids or solids in that they will ignite only after a longer period of time or when in large amounts.
Substances and mixtures which, in contact with water, emit flammable gases	As the class name suggests, these products react with water to release flammable gases. In some cases, the flammable gases may ignite very quickly (spontaneously).
Organic peroxides	These products are unstable, highly reactive, or explosive. They may cause a fire or explosion if heated.
Corrosive to metals	These products may be corrosive (chemically damaging or destructive) to metals.
Combustible dusts	This class is used to warn of products that are finely divided solid particles. If dispersed in air, the particles may catch fire or explode if ignited.

Physical hazard class	General description
Simple asphyxiants	These products are gases that may displace (take the place of) oxygen in air and cause rapid suffocation.
Physical hazards not otherwise classified	This class is meant to cover any physical hazards that are not covered in any other physical hazard class. These hazards involve chemical reactions that result in serious injuries or deaths when the reactions occur. If a product is classified in this class, the hazard statement on the label and SDS will describe the nature of the hazard.

Table 3. Overview of health hazard classes

Health hazard class	General description
Acute toxicity	These products are fatal, toxic (poisonous), or harmful if they are inhaled (breathed in), if they come into contact with skin, or if they are ingested (swallowed). "Acute toxicity" refers to effects that occur following: • Skin contact or ingestion exposure to: – A single dose, or – Multiple doses given within 24 hours • An inhalation exposure of 4 hours Acute toxicity could result from exposure to the product itself. It could also result from a product that, upon contact with water, releases a gas that can cause acute toxicity.
Skin corrosion /irritation	This class covers products that cause severe skin burns (corrosion) and products that cause skin irritation.
Serious eye damage /eye irritation	This class covers products that cause serious eye damage (corrosion) and products that cause eye irritation.
Respiratory or skin sensitization	A respiratory sensitizer is a product that may cause allergy or asthma symptoms or breathing difficulties if inhaled. A skin sensitizer is a product that may cause an allergic skin reaction.
Germ cell mutagenicity	This hazard class includes products that may cause or are suspected of causing genetic defects. Genetic defects are permanent changes (mutations) to body cells that can be passed on to future generations.
Carcinogenicity	This hazard class includes products that cause or are suspected of causing cancer.

Health hazard class	General description
Reproductive toxicity	This hazard class includes products that may damage or are suspected of damaging fertility (the ability to conceive children) or the embryo, fetus, or offspring. Note: There is also a category that includes products that may cause harm to breast-fed children.
Specific target organ toxicity – single exposure	This hazard class covers products that cause or may cause damage to organs (for example, the liver, kidneys, or blood) after a single exposure. This class also includes a category for products that cause respiratory irritation, drowsiness, or dizziness.
Specific target organ toxicity – repeated exposure	This hazard class covers products that cause or may cause damage to organs (for example, the liver, kidneys, or blood) following prolonged or repeated exposure.
Aspiration hazard	This hazard class is for products that may be fatal if they are swallowed and enter the airways.
Biohazardous infectious materials	These materials are micro-organisms (e.g., viruses, bacteria, or fungi), nucleic acids (e.g., DNA or RNA), or proteins that cause or are probable causes of infection, with or without toxicity, in humans or animals.
Health hazards not otherwise classified	This class covers products that are not included in any other health hazard class. These hazards occur following acute or repeated exposure. They have adverse effects on the health of a person exposed to them — including injury or death. If a product is classified in this class, the hazard statement will describe the nature of the hazard.

Hazard categories

Each hazard class contains at least one category. The hazard categories are assigned a number (1, 2, etc.). Categories may also be called "types." Types are assigned an alphabetical letter (A, B, etc.). In a few cases, subcategories are also specified. Subcategories are identified with a number and a letter (for example, 1A and 1B).

Some hazard classes have only one category (for example, "Corrosive to metals"). Others may have two categories (for example, "Carcinogenicity" [cancer]) or three categories (for example, "Oxidizing liquids"). There are a few hazard classes with five or more categories (for example, "Organic peroxides").

The category tells you how hazardous the product is (that is, the severity of hazard).

- Category 1 is always the greatest level of hazard. (In other words, it is the most hazardous within that class.) If Category 1 is further divided, Category 1A within the same hazard class is a greater hazard than Category 1B.
- Category 2 within the same hazard class is more hazardous than Category 3, and so on.

There are a few exceptions to this rule. For example, for the "Gases under pressure" hazard class, the hazard categories are "Compressed gas," "Liquefied gas," "Refrigerated liquefied gas," and "Dissolved gas." These classes relate to the physical state of the gas when packaged. They do not describe the degree of hazard.

Hazard category	Level of hazard
1	More hazardous
2A	
2B	
3	Less hazardous

Also, the "Reproductive toxicity" hazard class has a separate category called "Effects on or via lactation." "Effects on or via lactation" was not assigned a numbered category. Reproductive toxicity also has categories 1 and 2, which relate to effects on fertility and/or the embryo, fetus, or offspring. "Effects on or via lactation" is considered a different, but related, hazard within the "Reproductive toxicity" class.

Products not covered by WHMIS 2015

The following types of products are not covered under WHMIS 2015:

- Explosives
- Cosmetics, devices, drugs, and foods
- Pest control products (pesticides)
- Consumer products (for example, cleaning products, adhesives, and lubricants)
- Wood or products made of wood
- Nuclear (radioactive) substances
- Hazardous waste
- Tobacco and tobacco products
- Manufactured articles